Evidence-Grounded Reasoning Agents for Human-AI Decision Support

Overview Decision makers often need to reconcile long reports, partial observations, and successive updates describing the same evolving situation. Different sources may emphasize different events, report inconsistent times, or supersede an earlier account. Generative AI can summarize these materials, but a concise narrative may hide missing evidence, temporal inconsistencies, or unsupported inferences. Moreover, a seemingly clear explanation can encourage excessive reliance on an incorrect AI assessment. Trustworthy human-AI decision support requires both an accurate internal representation of events and a means for people to inspect and challenge the evidence behind conclusions.

We propose evidence-grounded reasoning agents that maintain structured representations of evolving reports and support interactive verification. The main challenges are: (1) extracting and updating consistent event knowledge across long and heterogeneous documents; (2) reasoning over temporal, causal, and provenance relations while preserving uncertainty; and (3) designing interactions that improve decision quality rather than simply increasing confidence in an AI adviser. Our NSF CAREER work on event-graph extraction and our research on faithful reasoning provide the methodological foundation.

We will pursue three interrelated research threads:

- **Dynamic and consistent event knowledge:** We will extract events, participants, attributes, and temporal or causal links from long document collections using open-schema, question-answer-based representations. We will connect related observations across documents, preserve provenance and conflicting descriptions, and develop incremental update methods that distinguish genuine changes from inconsistent reports. Global constraints will flag impossible sequences and relationships without silently deleting uncertain information.
- **Multi-step inference grounded in changing evidence:** We will integrate event knowledge with retrieved passages to answer questions requiring reasoning across sources and time. Each inference will carry its evidential dependencies, distinguishing direct observations, derived relationships, and assumptions. Stepwise verification will identify contradictions and missing support; when reports conflict, the agent will represent alternative explanations rather than force an unjustified single answer.
- **Interactive evidence inspection for better human decisions:** We will enable users to query why a conclusion was reached, inspect relevant passages, challenge a specific step, and request additional verification. Agents will revise their assessments when updates arrive or a user reveals a missing constraint. We will investigate presentation strategies for provenance and uncertainty, explicitly testing whether they decrease harmful reliance on incorrect recommendations while preserving appropriate reliance on correct ones.

Keywords: Human-AI decision support, Event graphs, Evidence provenance, Temporal reasoning.

Intellectual Merit The project will combine consistent event representation, provenance-preserving reasoning, and human-centered verification into a coherent decision-support framework. Evaluation will cover event-graph accuracy, temporal consistency, update robustness, faithfulness of explanations, and the quality and cost of decisions in controlled user studies, with flat summarization and standard retrieval-augmented assistants as baselines.

Naval Relevance and Broader Impact The work is relevant to Naval command decision making and situation understanding when reports arrive over time and users must evaluate the evidential basis for an assessment. We will use unclassified document collections and controlled information-update scenarios and disseminate reusable evaluation resources.